

Scalability & Future Plan

Date	30 June 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	1 Mark

Scalability & Future Plan

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static dataset	Limited real-time prediction	High
2	Local deployment	Limited user access	Medium
3	No live weather integration	Lower prediction accuracy	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports local users	Deploy on cloud with load balancing
2	Data Storage	Static dataset	Integrate cloud database and live data
3	Performance	Local Flask server	Optimize APIs and enable caching
4	Security	Basic application	Add user authentication and HTTPS

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Live Weather API Integration	1–2 Months	Real-time predictions
Phase 2	Interactive Flood Risk Maps (GIS)	2–3 Months	Better visualization
Phase 3	SMS / Email Alert System	3–4 Months	Early warning notifications
Phase 4	Cloud Deployment & Mobile App	4–6 Months	Improved accessibility and scalability